Handwritten Digit Recognition System

System Architecture Specification & Technical Documentation — Category: Computer Vision & Deep Learning (CNN)

Author: Rakshitha S (Full-Stack & AI/ML)
Institution: Jnana Vikas Inst of Tech, Bengaluru
Repository: https://github.com/rakshapatil526/Handwritten-Digit-Recognition-System

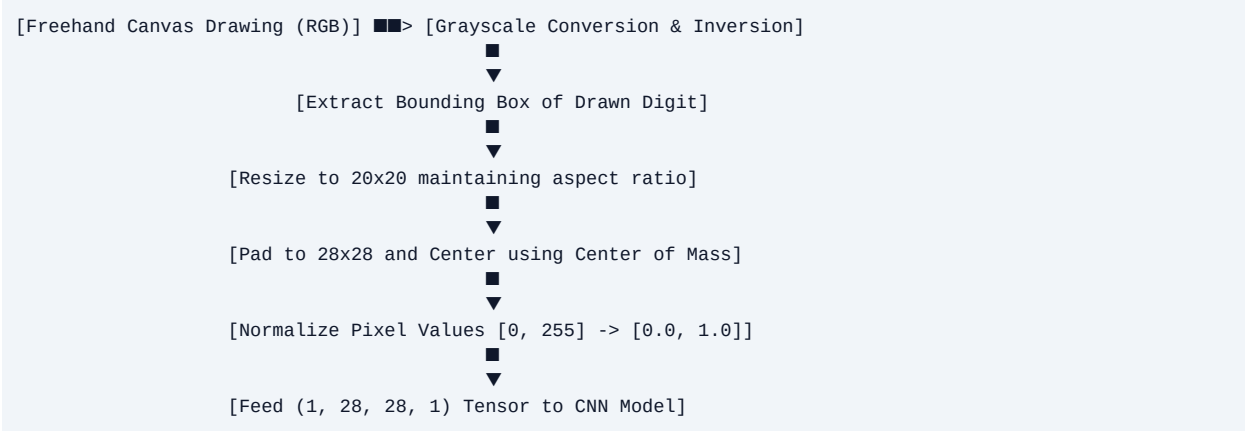
Status: Verified Repository
Doc ID: SPEC-HANDWRITTEN-DIGIT-RECOGNITION
Date: March 2026

---

1. Computer Vision Problem & MNIST Benchmark

Recognizing handwritten numeric characters is a foundational computer vision challenge due to variations in handwriting slant, stroke thickness, and drawing speed. This application provides an interactive deep learning platform powered by a Convolutional Neural Network (CNN) trained on the 70,000-image MNIST benchmark dataset (Yann LeCun).

2. Image Standardization Pipeline



3. Convolutional Neural Network Architecture

Layer Type	Parameters / Shape	Activation & Purpose
Conv2D (Block 1)	32 filters, 3x3 kernel	ReLU + BatchNormalization (Edge & stroke detection)
Conv2D (Block 1)	32 filters, 3x3 kernel	ReLU + BatchNormalization
MaxPooling2D	Pool size 2x2, Dropout 0.25	Spatial dimension reduction (14x14) & regularization
Conv2D (Block 2)	64 filters, 3x3 kernel	ReLU + BatchNormalization (High-level digit shapes)
Conv2D (Block 2)	64 filters, 3x3 kernel	ReLU + BatchNormalization
MaxPooling2D	Pool size 2x2, Dropout 0.25	Spatial dimension reduction (7x7)
Flatten & Dense	Dense 256 units, Dropout 0.5	ReLU fully connected feature combination
Output Dense	Dense 10 units	Softmax probability distribution over digits 0 through 9

4. Performance & Interactive Interface

• **Evaluation:** Test set accuracy of ~99.1% on hold-out MNIST test samples; Plotly training curves show robust convergence.

• **Frontend:** Streamlit with `streamlit-drawable-canvas` enabling live touch/mouse digit drawing, clear canvas buttons, and real-time probability bar charts.